Enterprise Network Design Portfolio Project

Prepared for portfolio purposes by Isaac. Proposed designs and illustrative scenarios are clearly distinguished from verified work experience.

Executive Summary

Detailed conceptual network for a three-floor office supporting 150 users.

Scope and Assumptions

Illustrative design sample only; not claimed as a live deployment.

Logical Topology

Internet > HA Firewalls > Core Switch Stack > Distribution > Access Layer.

Layer	Technology	Purpose
Core	L3 Switching	Routing
Access	PoE Switches	Endpoints

IP Plan

Illustrative VLAN strategy.

VLAN	Department	Subnet
10	Mgmt	10.10.10.0/24
20	HR	10.10.20.0/24
30	Finance	10.10.30.0/24
40	Ops	10.10.40.0/24
50	ICT	10.10.50.0/24

Floor 1

Reception and administration network.

Floor 2

Finance and operations network.

Floor 3

ICT, training room and server access.

Wireless Design

Multiple Wi■Fi 6 APs with roaming.

Security Controls

802.1X, MFA, NAC, IDS/IPS, SIEM integration.

Control	Purpose
MFA	Identity
IDS/IPS	Threat detection
Backups	Recovery

Risk Assessment

Operational risks and mitigations.

Risk	Impact	Mitigation
ISP outage	High	Secondary ISP
Power loss	High	UPS

Monitoring

SNMP, Syslog and alerting workflows.

Troubleshooting

Layered model from L1 to L7.